

PTVA's MULUND COLLEGE OF COMMERCE (AUTONOMOUS)			
Programme: B.Sc. Information Technology		Class: TY	Semester: V (AUTO)
Exam: ESE (ATKT)	Name of the Course: Internet of Things		
Date of Exam: 02/12/2025	Marks: 60	Time: 2 hours	Autonomous
Instructions: (1) <u>All</u> questions are <u>compulsory</u> . (2) Make <u>suitable assumptions</u> wherever necessary and <u>state the assumptions</u> made. (3) Answers to the <u>same question</u> must be <u>written together</u> . (4) Numbers to the <u>right</u> indicate <u>marks</u> . (5) Draw <u>neat labeled diagrams</u> wherever <u>necessary</u> . (6) Use of non-programmable calculator is <u>allowed</u> .			

Q 1 Attempt any TWO of the following: 12

- A. Describe the application of IOT in various areas.
- B. Explain the concept of magic as metaphor with the help of John McKerrell's WhereDial.
- C. What do you mean by first class citizen of the Internet? Explain.
- D. What is a MAC address? What is its use? How does it differ from IP address?

Q 2 Attempt any TWO of the following: 12

- A. What are the qualities of being a good IoT citizen?
- B. What are the merits and demerits of mixing open source and close source?
- C. With the help of an example explain the process of Scaling up the electronics.
- D. Explain the hardware and openness with respect to raspberry Pi.

Q 3 Attempt any TWO of the following: 12

- A. Discuss the Nik O Larry's ambient orb.
- B. What is 3D printing? What are its uses?
- C. What is comet? Explain.
- D. Discuss the HTML5 WebSockets in detail.

Q 4 Attempt any TWO of the following: 12

- A. Explain in detail the process of debugging the code for embedded devices.
- B. How do we write the code for improving performance and battery life of embedded devices? Explain.
- C. What are the constraints while receiving funding for IoT start-ups?
- D. What is crowd funding? Explain.

Q 5 Attempt any TWO of the following: 12

- A. Write a short note on mass-producing the case and other fixtures.
- B. What is the important of certification? Why it is required?
- C. "The download spiral of mankind from a better state." Discuss.
- D. Discuss the human cost associated with the production of IOT devices.